



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to OM SAI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

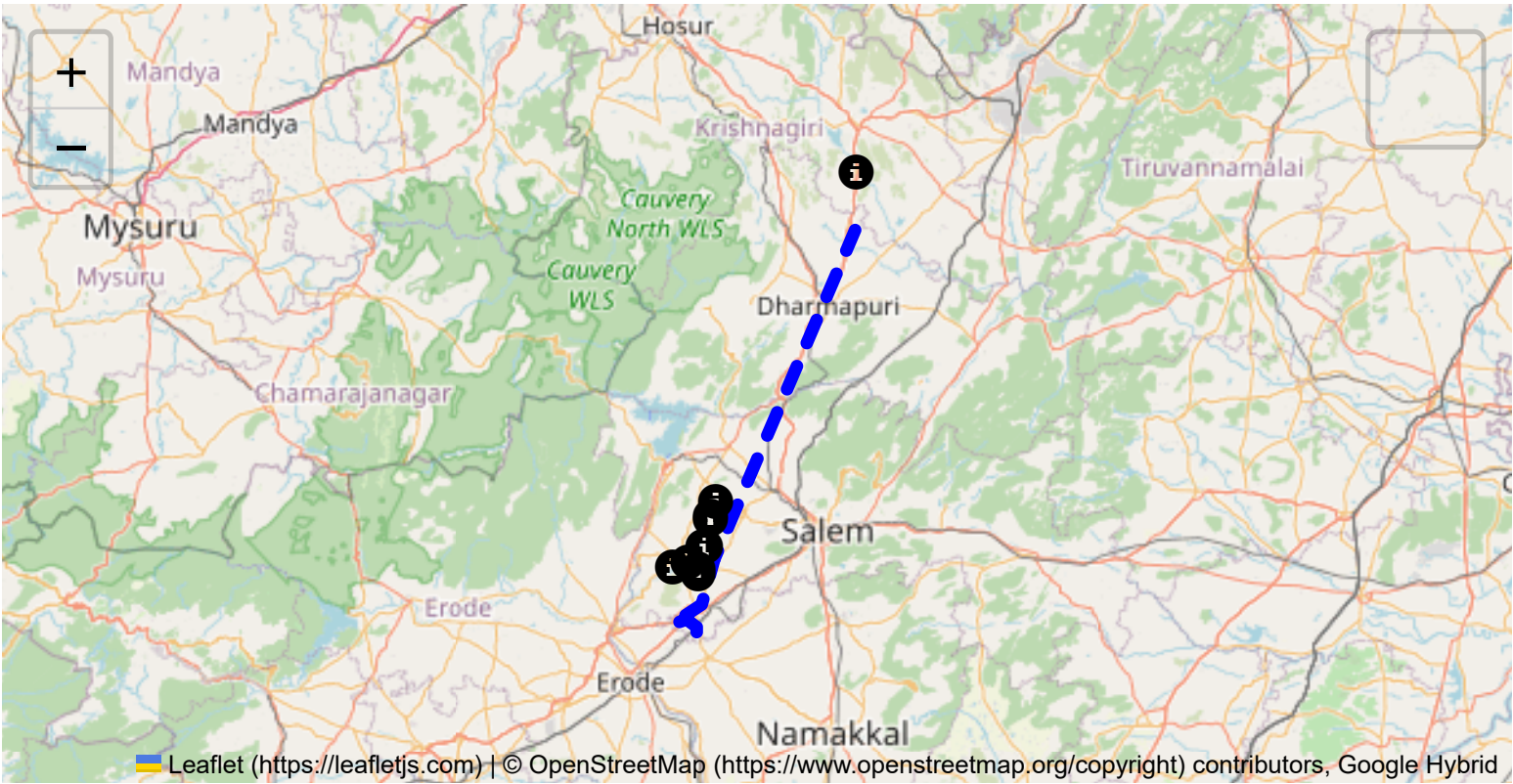
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

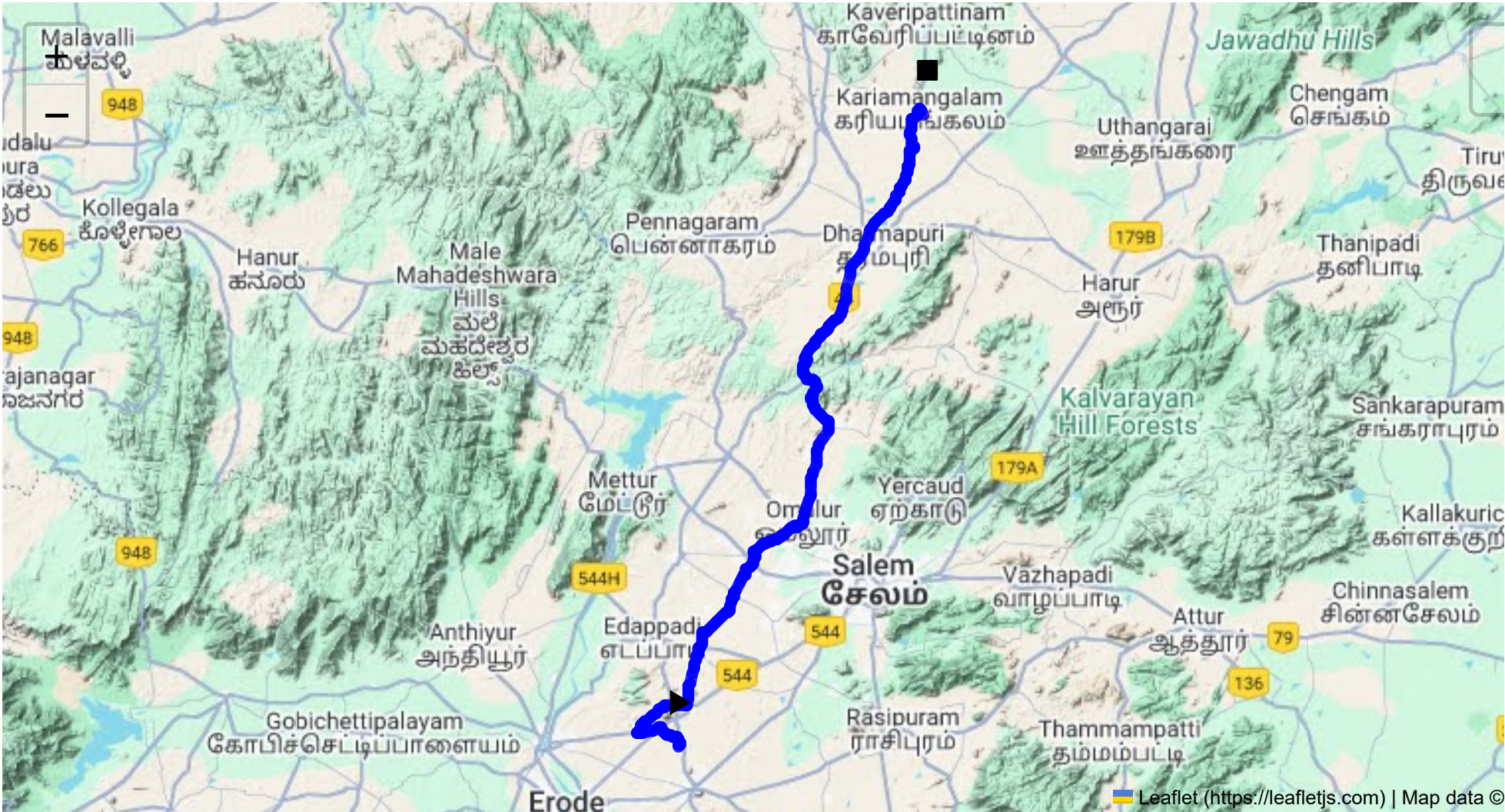
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 126.92 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (12.298155, 78.216146)



Welcome to the Journey Risk Management Study

1. Route Overview: The route from CVQF+23W, Sangagiri to Karimangalam follows a network of state highways and local roads passing through multiple rural and semi-urban areas. Major waypoints include Sankari RS, Sankari West, Padaiveedu, Erumaipatti, and Koranampatti. These areas suggest a mix of narrow rural roads intertwining with busier state highways, possibly requiring careful navigation especially through village junctions.

2. Weather Conditions: Typical weather in Tamil Nadu includes a hot and dry climate with heavy rains during the monsoon season (June-September). Potential weather-related hazards include heavy rain leading to reduced visibility and slippery roads, as well as the possibility of flooding in low-lying areas. Heat can also affect vehicle performance.

3. Traffic Patterns: The route typically experiences moderate traffic, with peak congestion during morning (8-10 AM) and evening (5-7 PM) hours, particularly near market towns and educational institutions. Sankari and Erumaipatti can experience delays during these times due to local activities.

4. Road Quality and Infrastructure: While state highways (like SH 86) are maintained and generally in decent condition, local segments might encounter potholes, uneven surfaces, or worn road markings. The quality diminishes as routes transition into less urbanized regions. Bridge conditions, especially following heavy rains, should be monitored.

5. Alternative Routes: In case of emergencies, one possible alternative might be routing through major highways like NH44, albeit this would increase travel distance and time. Regular checks on road closures due to weather or construction are advised.

6. Local Regulations: Transporting hazardous materials requires compliance with state regulations, including permits, signage, and specific driving hours. Tamil Nadu has strict enforcement regarding such cargo, particularly in populated areas.

7. Historical Incidents: While precise data might not be readily available, routes involving rural and semi-urban areas typically see incidents related to sharp turns, overloading, and inattention during night driving. Historical weather events have also caused local disruptions.

8. Environmental Considerations: The route does not traverse ecologically sensitive zones in a significant way, but care should be taken near agricultural areas to prevent contamination from hazardous leaks. Local bodies might prohibit parking near water bodies and farms.

9. Communication Coverage: Most of the route is covered by major cellular networks but dead zones may be encountered in sparsely populated rural stretches. Reliable communication might diminish in pockets around Erumaipatti and Koranampatti.

10. Emergency Response Times: In case of emergencies, response time may vary from 30 minutes to over an hour depending on proximity to the nearest town. Erumaipatti is likely to offer the fastest access to emergency services due to its relative size and facilities.

12. Overall Risk Assessment: Overall, the route poses moderate risk primarily due to variable road conditions and potential weather challenges. Preparedness involving up-to-date communication equipment, adherence to scheduled transportation hours, and contingency plans for inclement weather and road quality are advisable. Drivers should be trained to handle vehicular issues in rural settings and must maintain alertness particularly during peak hours and night driving.

By considering these factors, drivers can better prepare for a smoother and safer journey along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Turn	Medium	11.46916, 77.82431	30 KM/Hr
10	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
11	Turn	High	11.49465, 77.88488	15 KM/Hr
12	Turn	Medium	11.56096, 77.89931	30 KM/Hr
13	Turn	Medium	11.57649, 77.90610	30 KM/Hr
14	Turn	Medium	11.59214, 77.90656	30 KM/Hr
15	Turn	Medium	11.63833, 77.94818	30 KM/Hr
16	Turn	Medium	11.64171, 77.95184	30 KM/Hr
17	Turn	Medium	11.64243, 77.95198	30 KM/Hr
18	Turn	Medium	11.64721, 77.95118	30 KM/Hr
19	Turn	High	11.67870, 77.96846	15 KM/Hr
20	Turn	High	11.70227, 77.97864	15 KM/Hr
21	Turn	Medium	11.71715, 77.99969	30 KM/Hr
22	Turn	Medium	11.71921, 78.01133	30 KM/Hr
23	Turn	Medium	11.72737, 78.02248	30 KM/Hr
24	Turn	Medium	11.73775, 78.03465	30 KM/Hr
25	Turn	High	11.73651, 78.03878	15 KM/Hr
26	Turn	Medium	11.74189, 78.04660	30 KM/Hr
27	Turn	Medium	11.74309, 78.04725	30 KM/Hr
28	Turn	Medium	11.78039, 78.05792	30 KM/Hr
29	Turn	Medium	11.86753, 78.08331	30 KM/Hr
30	Turn	Medium	11.86981, 78.08387	30 KM/Hr
31	Turn	Medium	11.88041, 78.08143	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
32	Turn	Medium	11.91302, 78.05856	30 KM/Hr
33	Turn	Medium	11.95625, 78.04748	30 KM/Hr
34	Turn	High	11.95824, 78.04981	15 KM/Hr
35	Turn	High	11.96040, 78.04947	15 KM/Hr
36	Turn	Medium	11.96320, 78.05139	30 KM/Hr
37	Turn	Medium	11.96401, 78.05134	30 KM/Hr
38	Turn	Medium	11.96463, 78.05077	30 KM/Hr
39	Turn	Medium	11.96504, 78.04975	30 KM/Hr
40	Turn	High	11.96597, 78.04920	15 KM/Hr
41	Turn	Medium	11.96679, 78.04942	30 KM/Hr
42	Turn	Medium	11.96804, 78.05080	30 KM/Hr
43	Turn	Medium	12.24611, 78.19519	30 KM/Hr
44	Turn	High	12.30425, 78.20845	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
29	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
31	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59214, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67870, 77.96846



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74189, 78.04660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74309, 78.04725



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78039, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95625, 78.04748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95824, 78.04981



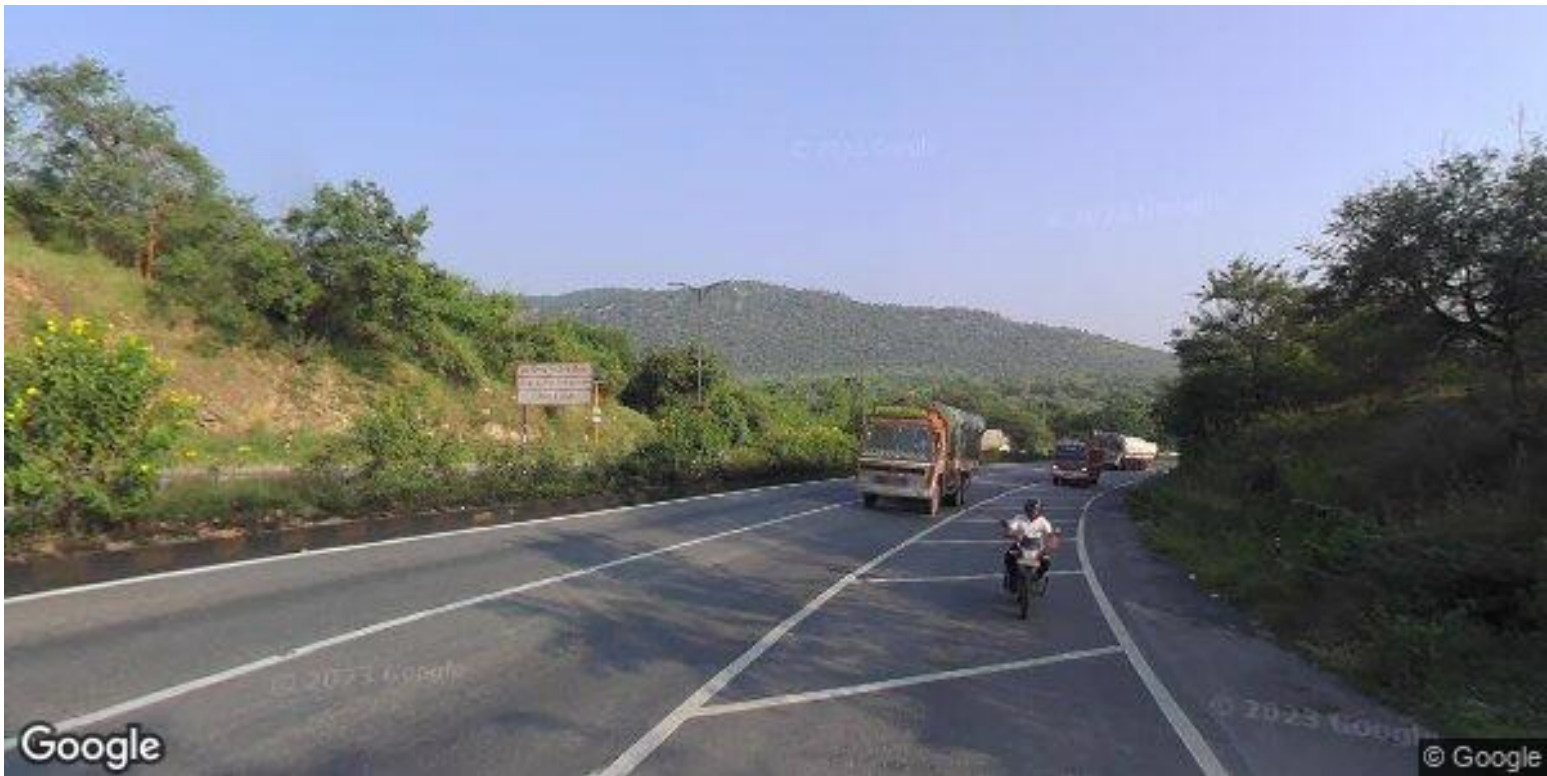
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96463, 78.05077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96679, 78.04942



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24611, 78.19519



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.30425, 78.20845